

0.1 Section 8.1 - Generating Functions

How do we determine an explicit formula for a recursive sequence?

For example,

$$\begin{aligned}f_0 &= 0 \\f_1 &= 1 \\f_{n+1} &= f_n + f_{n-1}, \quad n \geq 1\end{aligned}$$

Has an explicit function of

$$f_n = \frac{1}{\sqrt{5}} \left(\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right)$$

Let's consider a nicer example

$$\begin{aligned}a_0 &= 6 \\a_1 &= 13 \\a_n &= 5a_{n-1} - 6a_{n-2}, \quad n \geq 2\end{aligned}$$

Let's write out the first few terms

$$\begin{aligned}a_2 &= 5a_1 - 6a_0 \\&= 5 \cdot 13 - 6 \cdot 5 \\&= 29\end{aligned}$$

$$\begin{aligned}a_3 &= 5 \cdot 29 - 6 \cdot 13 \\&= 67\end{aligned}$$

$$\begin{aligned}a_4 &= 5 \cdot 67 - 6 \cdot 29 \\&= 161\end{aligned}$$

We form a **generating function** for the sequence

$$a_0, a_1, a_2, \dots$$

Which is a power series whose coefficients are the sequence

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots$$

From our example above, the generating function is

$$f(x) = 6 + 13x + 29x^2 + 67x^3 + 161x^4 + \dots$$

To reduce this, let's go back to our recursive function

$$a_n = 5a_{n-1} - 6a_{n-2}$$

Rearranging, we get

$$a_n - 5a_{n-1} + 6a_{n-2} = 0$$

Now let's bring in our generating function

$$\begin{aligned} f(x) &= 6 + 13x + \cancel{29x^2} + \cancel{67x^3} + \cancel{161x^4} + \dots \\ -5xf(x) &= -30x - \cancel{65x^2} - \cancel{145x^3} - \cancel{335x^4} + \dots \\ 6x^2f(x) &= \quad \quad \cancel{36x^2} + \cancel{78x^3} + \cancel{174x^4} + \dots \end{aligned}$$

$$\begin{aligned} (1 - 5x + 6x^2)f(x) &= 6 - 17x \\ f(x) &= \frac{6 - 17x}{1 - 5x + 6x^2} \\ &= \frac{6 - 17x}{(1 - 2x)(1 - 3x)} \\ &= \frac{A}{1 - 2x} + \frac{B}{1 - 3x} \end{aligned}$$

$$A(1 - 3x) + B(1 - 2x) = 6 - 17x$$

$$\begin{aligned}\text{Let } x &= \frac{1}{3} \\ B(1 - \frac{2}{3}) &= 6 - \frac{17}{3} \\ B \cdot \frac{1}{3} &= \frac{18}{3} - \frac{17}{3} \\ \boxed{B} &= 1\end{aligned}$$

$$\begin{aligned}\text{Let } x &= \frac{1}{2} \\ A(1 - \frac{3}{2}) &= 6 - \frac{17}{2} \\ A(-\frac{1}{2}) &= \frac{12}{2} - \frac{17}{2} \\ -\frac{1}{2}A &= -\frac{5}{2} \\ \boxed{A} &= 5\end{aligned}$$

So

$$\boxed{f(x) = \frac{5}{1-2x} + \frac{1}{1-3x}}$$

Recall 1

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots = \sum_{n=0}^{\infty} x^n$$

So

$$\begin{aligned}f(x) &= 5 \sum_{n=0}^{\infty} (2x)^n + \sum_{n=0}^{\infty} (3x)^n \\ \sum_{n=0}^{\infty} a_n x^n &= \sum_{n=0}^{\infty} (5 \cdot 2^n + 3^n) x^n \\ a_n &= 5 \cdot 2^n + 3^n\end{aligned}$$

But wait. If

$$\begin{aligned}a_n &= 5a_{n-1} - 6a_{n-2} \\a_n - 5a_{n-1} + 6a_{n-2} &= 0 \\ \implies x^2 - 5x + 6 &= 0 \\ (x-3)(x-2) &= 0 \\ x &= 2, 3\end{aligned}$$

But

$$a_n = 5 \cdot 2^n + 1 \cdot 3^n$$

How do we get the 5 and 1 (the coefficients)?

Using this shortcut, we can see that

$$a_n = A \cdot 2^n + B \cdot 3^n$$

Let's use $a_0 = 6$ and $a_1 = 13$

$$\begin{aligned}a_0 = 6 &= A \cdot 2^0 + B \cdot 3^0 \\ 6 &= A + B \\ A &= 6 - B\end{aligned}$$

$$\begin{aligned}a_1 = 13 &= A \cdot 2 + B \cdot 3 \\ 13 &= 2A + 3B \\ 13 &= 2(6 - B) + 3B \\ 13 &= 12 - 2B + 3B\end{aligned}$$

$$\boxed{B = 1}$$

$$A = 6 - 1$$

$$\boxed{A = 5}$$